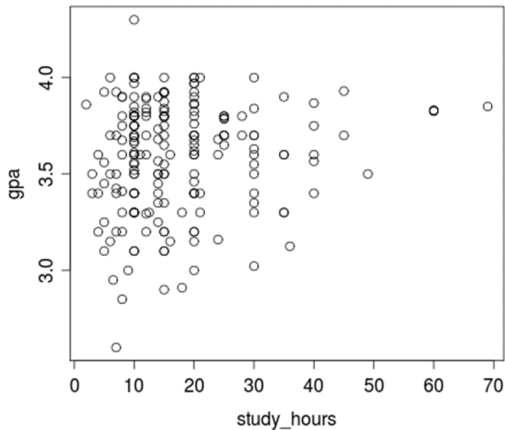


2.4 Two Variable Relationships: Two Quantitative Variables

- Goal: We need to turn “raw” data into information.
- Two quantitative variables
 - 1 Visualization: scatterplot
 - 2 Summary statistic: correlation

Scatterplot

- A **scatterplot** is a plot of pairs of quantitative values as points on a coordinate axis, showing the relationship between **two variables** measured on the **same cases**.



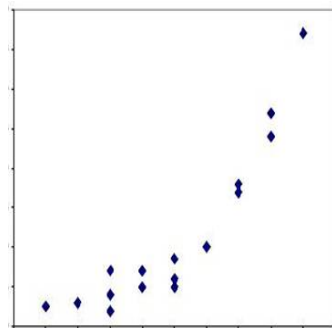
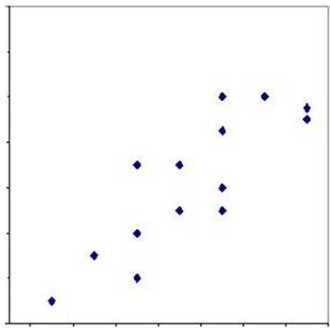
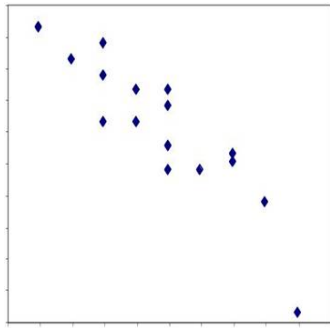
Source: Lock⁵ (2021), John Wiley & Sons, Inc.

Scatterplot

- If the objective is to predict a response variable Y from an explanatory variable X , then by convention, we select Y as the vertical axis variable and X as the horizontal axis variable.
- Information from scatterplots
 - 1 **Type** of association: linear or nonlinear
 - 2 **Direction** of association: positive (direct) or negative (inverse)
 - 3 **Strength** of association: weak, moderate, or strong
 - 4 **Outliers** (individual values that fall outside the overall pattern of the relationship)

Type of Association

- Two variables have a **linear** relationship if the points in a scatterplot roughly follow a **straight** line with a **non-zero slope**.

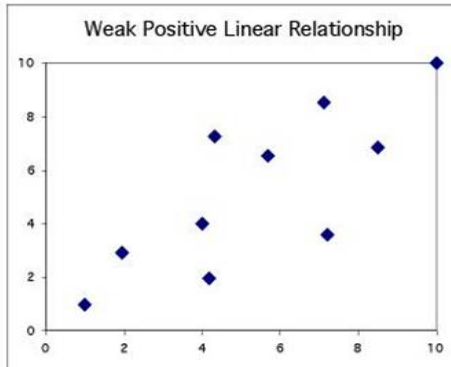
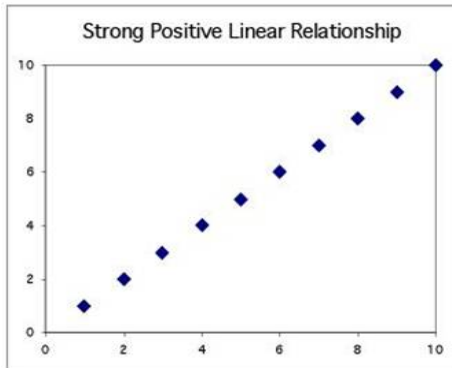


Direction of Association

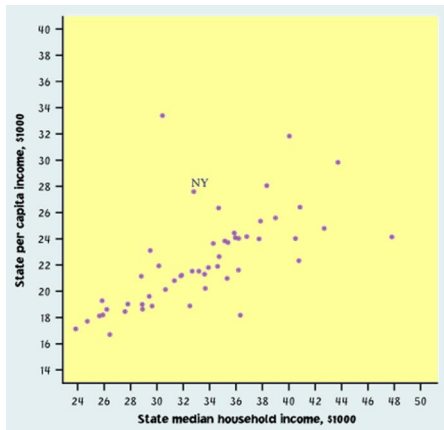
- Suppose **large** values of Y tend to have **large** values of X , and **small** values on X also tend to have **small** values on Y . This is a **positive association** or **direct relationship**.
 - Example: Height and weight – taller people tend to be heavier.
- Suppose **large** values of X tend to have **small** values of Y , and vice-versa. This is a **negative association** or **inverse relationship**.
 - Example: Smoking and life expectancy – people who smoke more cigarettes per day tend not to live as long as those who smoke fewer
- Two variables are **not associated** if knowing the value of one variable does not give you any information about the value of the other variable.
- Note: Statistical associations are overall tendencies.

Strength of Association

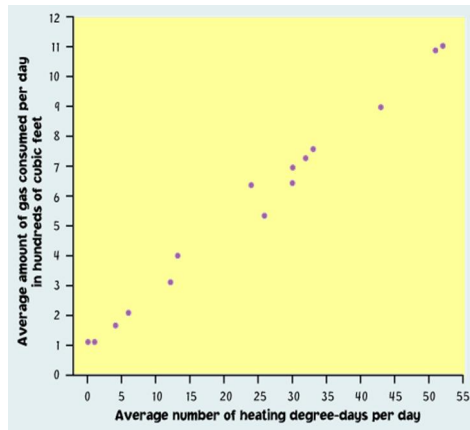
- **Strength** of a relationship is determined by how closely the points follow a clear form.



Example 2.13 | Strength of Association



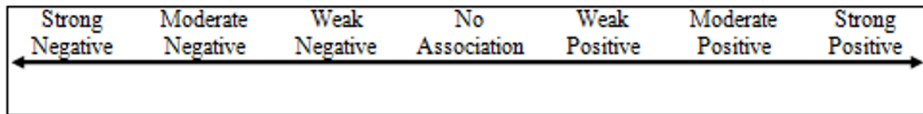
Note: This is a **weak** relationship. For a particular state median household income, you cannot predict the state per capita income very well.



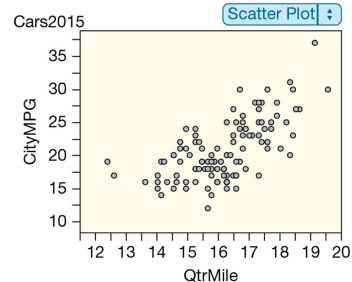
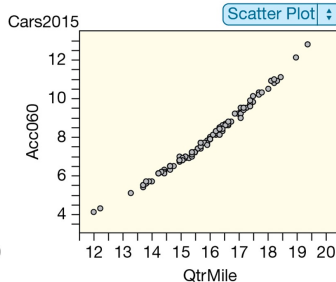
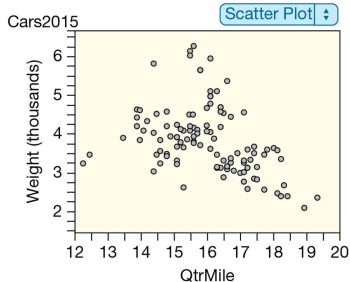
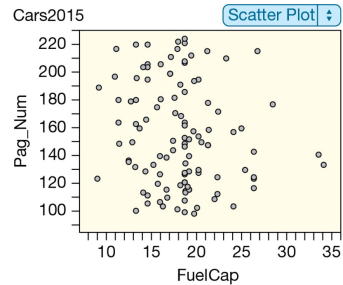
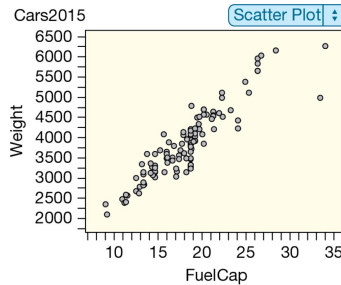
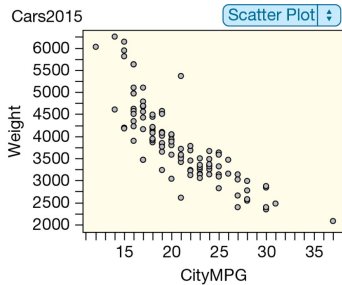
Note: This is a very **strong** relationship. The daily amount of gas consumed can be predicted quite accurately for a given temperature value.

Example 2.14 | Cars Data (Direction and Strength)

- Make initial guesses for the strength and direction of association for each of the following:
 - 1 Weight (in pounds) vs. CityMPG
 - 2 Weight (in pounds) vs. FuelCapacity (in gallons)
 - 3 PageNum (page number in Consumer Reports) vs. FuelCapacity (in gallons)
 - 4 Weight vs. QtrMile (time to go 1/4 mile in seconds)
 - 5 Acc060 (acceleration time from 0 to 60 mph in seconds) vs. QtrMile
 - 6 CityMPG vs. QtrMile

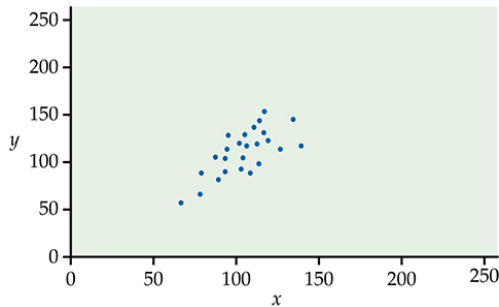
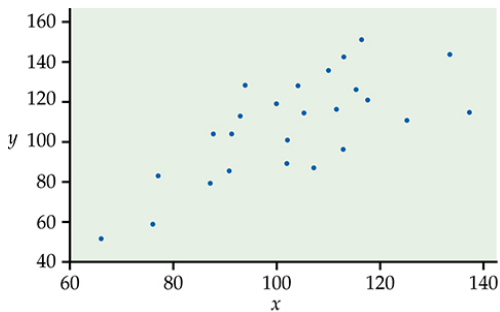


Example 2.14 | Cars Data (Direction and Strength)



Measure of Linear Relationship | Motivation

- Which one shows the stronger linear association? and why?



- Our eyes sometimes can be deceived. We need a [numerical measure of linear association](#) for quantifying the closeness of the scatter to a linear relation.

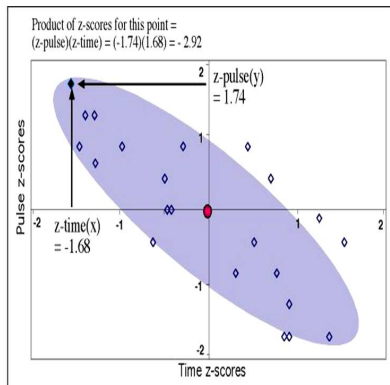
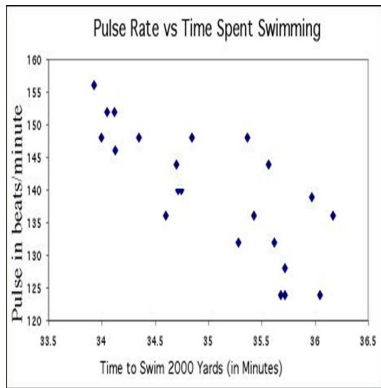
Measure of Linear Relationship | Correlation

- The **correlation** quantifies the **direction** and **strength** of the **linear relation** between **two numerical (quantitative) variables**.
- Sample correlation vs. Population correlation
 - 1 **Sample correlation** r
 - 2 **Population correlation** ρ ("rho")

Sample Correlation | Understanding

- r is an average of the products of the standardized values of X and Y :

$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right).$$



Sample Correlation | Computation

■ Computation of the sample correlation r :

- 1 There are data on variables X and Y from a sample of n subjects, $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$.
- 2 Obtain s_x and s_y , the sample standard deviations for X and Y variables:

$$s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}, \quad s_y = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n - 1}}.$$

- 3 Compute the **standardized observations** of x_i and y_i as follows:

$$\frac{x_i - \bar{x}}{s_x}, \quad \frac{y_i - \bar{y}}{s_y}.$$

- 4 Finally,

$$r = \frac{1}{n - 1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right).$$

Example 2.15 | Correlation computation

- We have the following $n = 5$ pairs of observations

$$(2, 5), (1, 3), (5, 6), (0, 2), (2, 4).$$

First, make a scatterplot of this bivariate data.

Example 2.15 (continued) | Correlation computation

- Compute the sample correlation r by following its definition.

S1: Use the following worksheet to compute $\bar{x}$, $\bar{y}$, s_x , and s_y .

	x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
	2	5	0	1	0	1
	1	3	-1	-1	1	1
	5	6	3	2	9	4
	0	2	-2	-2	4	4
	2	4	0	0	0	0
Total	10	20	0	0	14	10

S2: We compute

$$\bar{x} = \frac{\sum x_i}{n} = \frac{10}{5} = 2, \quad \bar{y} = \frac{\sum y_i}{n} = \frac{20}{5} = 4$$

$$s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}} = \sqrt{\frac{14}{5 - 1}} = \sqrt{3.5} = 1.87$$

$$s_y = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n - 1}} = \sqrt{\frac{10}{5 - 1}} = \sqrt{2.5} = 1.58$$

Example 2.15 (continued) | Correlation computation

■ (Continued)

S3: The following worksheet is made for the computation of r :

x	y	$\frac{x-\bar{x}}{s_x}$	$\frac{y-\bar{y}}{s_y}$	$\left(\frac{x-\bar{x}}{s_x}\right) \cdot \left(\frac{y-\bar{y}}{s_y}\right)$
2	5	0.000	0.632	0.000
1	3	-0.535	-0.632	0.338
5	6	1.604	1.266	2.031
0	2	-1.069	-1.266	1.353
2	4	0.000	0.000	0.000
Total		0	0	3.722

S4: Therefore the sample correlation r is

$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right) = \frac{1}{5-1} \times 3.722 = \frac{3.722}{4} = 0.93$$

- **Interpretation:** $r = 0.93$ is positive and large. This implies a strong association where both X and Y tend to be small or both tend to be large.

Sample Correlation | Property 1

■ The **sign** of r indicates the **direction** of a linear relation:

- 1 $r > 0$ if x and y have a **positive** association (**direct** relationship).
- 2 $r < 0$ if x and y have a **negative** association (**inverse** relationship).
- 3 Why? Recall

$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

Sample Correlation | Property 2

- The **magnitude** of r indicates the **strength** of a linear relation: a large value of $|r|$, a value close to 1 or -1, represents a **strong** linear relation.
 - 1 $r = 1$ if all pairs of values (x, y) lie exactly on a **straight** line with a **positive** slope.
 - 2 $r = -1$ if all pairs of values (x, y) lie exactly on a **straight** line with a **negative** slope.
 - 3 $r = 0$ if there is no general linear tendency for y values to either increase or decrease with x values.

Sample Correlation | Property 3

■ $-1 \leq r \leq 1$



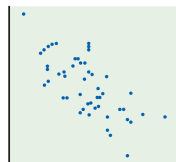
Correlation $r = 0$



Correlation $r = -0.3$



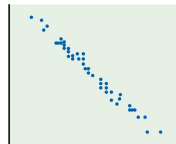
Correlation $r = 0.5$



Correlation $r = -0.7$

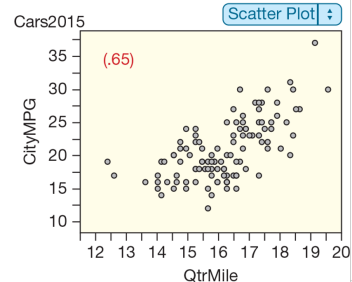
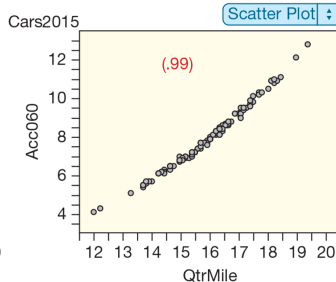
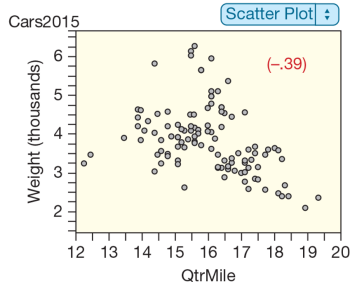
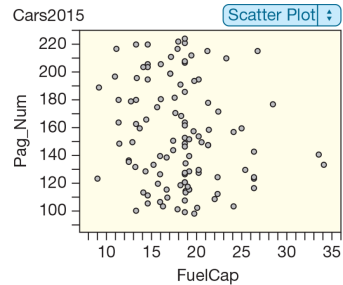
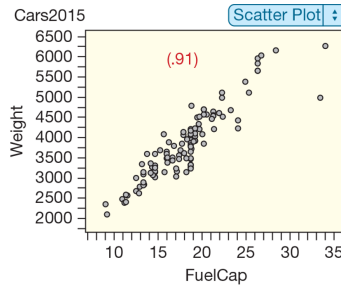
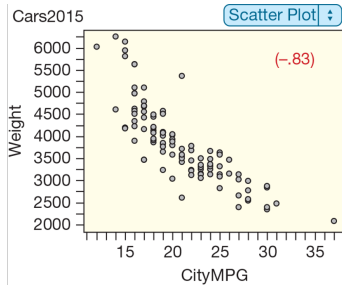


Correlation $r = 0.9$



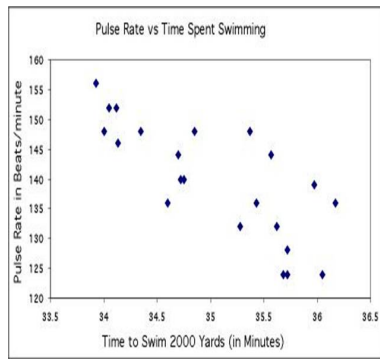
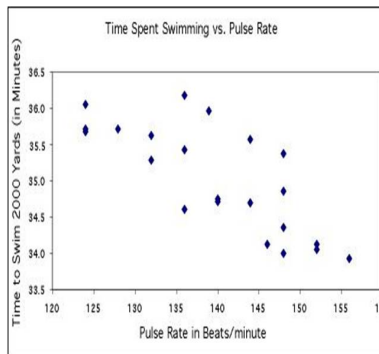
Correlation $r = -0.99$

Example 2.14 (revisited) | Cars Correlations



Sample Correlation | Property 4

- r does **not** distinguish X and Y .

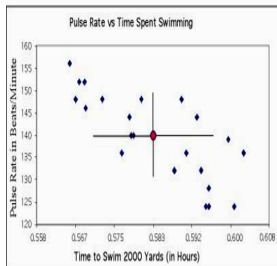
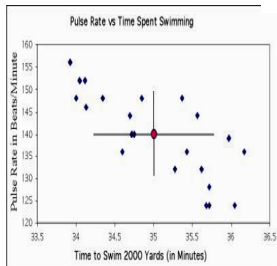


Note. “Time to swim” is the explanatory variable here, and belongs on the x-axis. However, in either plot r is the same ($r = -0.75$).

Sample Correlation | Property 5

■ r is **unitless**.

- 1 Changing the units of X and Y variables does **not** change the correlation coefficient r , because we get rid of all our units when we standardize (get z-scores).
- 2 Example. “Time to swim” is scaled on different units. However, in either plot r is the same ($r = -0.75$).

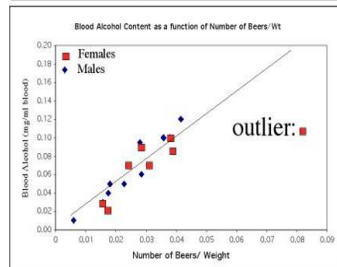
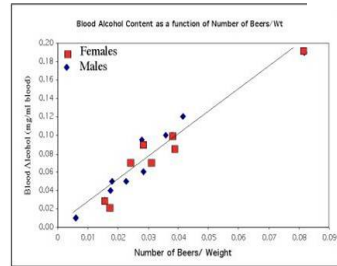


Sample Correlation | Property 6

- No matter how strong the association, r does **not** describe curved relationships.

Outliers in a Scatterplot

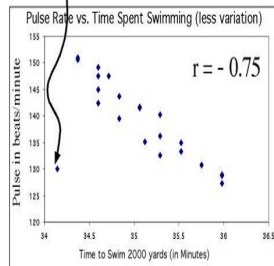
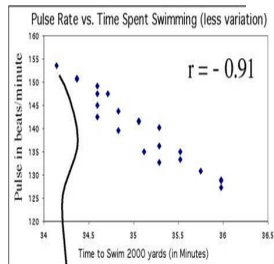
- The upper right-hand point here is **not** an outlier of the relationship. It is what you would expect for these many beers given the linear relationship between beers/weight and blood alcohol.
- The mid right-hand point is **not** in line with the others, so it **is** an outlier of the relationship.



Sample Correlation | Property 7

- Like the sample mean and sample standard deviation, the sample correlation is **not resistant**: r is strongly *affected* by a few *outliers*.

Note: Just moving one point away from the general trend here decreases the correlation from -0.91 (top) to -0.75 (bottom).

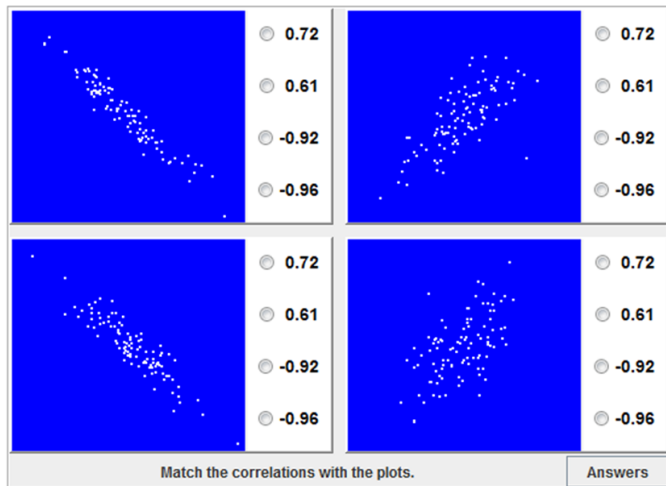


Correlation | Summary of Properties

- $-1 \leq r \leq 1$.
- The sign of r indicates the direction of association.
 - 1 Positive association: $r > 0$
 - 2 Negative association: $r < 0$
 - 3 No linear association: $r \approx 0$
- The closer r is to 1 or -1 , the stronger the linear association.
- r has no units and does not depend on the units of measurement.
- The correlation between X and Y is the same as the correlation between Y and X .
- r can be sensitive to outliers.
- Correlation does not imply causation!

Exercise 2.8 | Correlation Guessing Game

■ <https://www.guessthecorrelation.com/>



Source: Lock⁵ (2021), John Wiley & Sons, Inc.